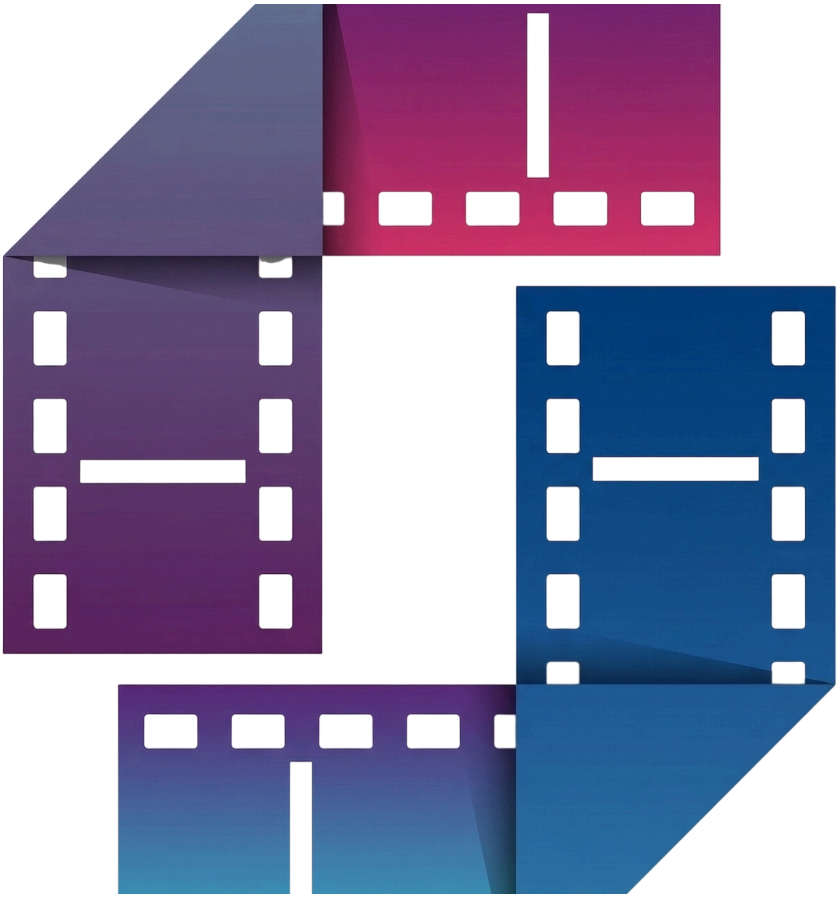
UnReel : Video Intelligence and Analysis Platform

- **Watch the intelligence unfold**



Explore UnReel: Live Analysis Demo

GitHub Link of UnReel: Video Intelligence and Analysis Platform

Introduction

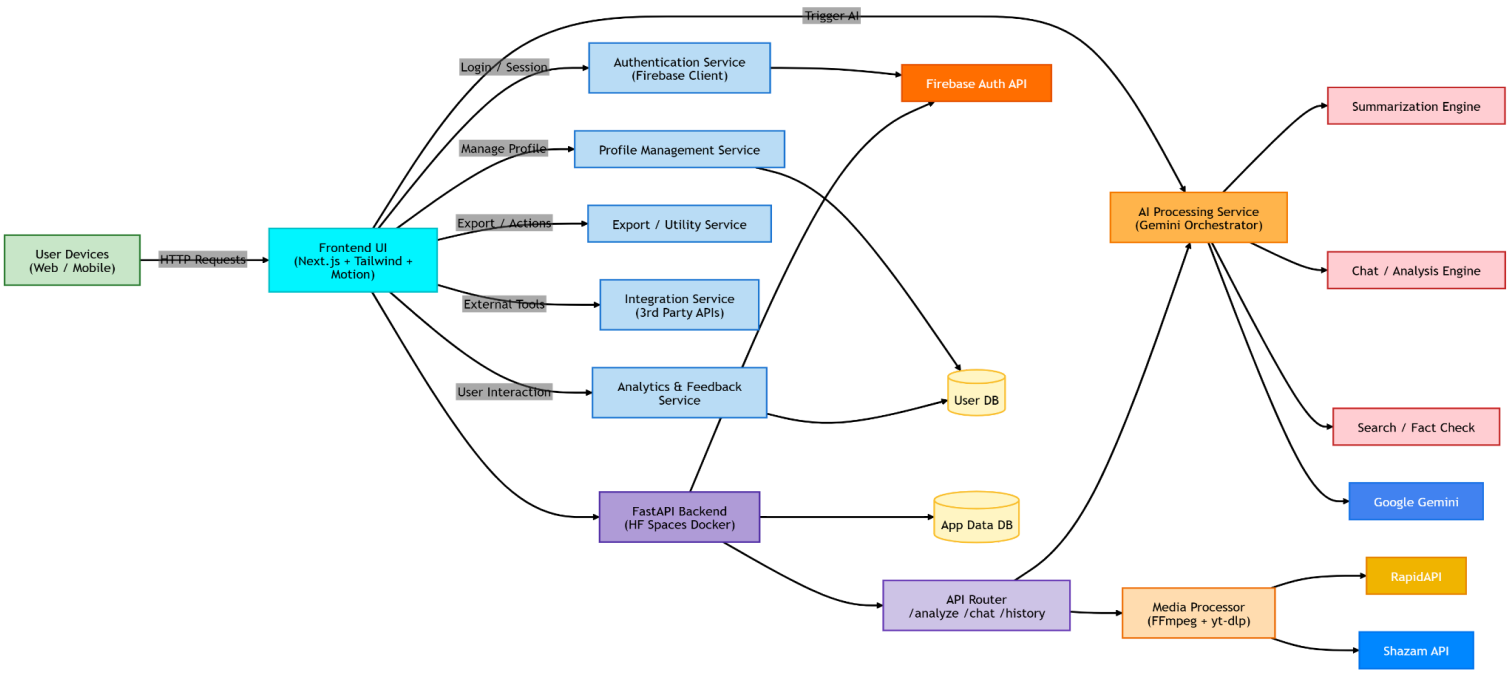
The **UnReel** AI-powered video intelligence application is a cutting-edge solution designed to enhance the efficiency of researchers, content creators, and digital investigators in managing and understanding short-form video content. Unlike traditional video players, this application leverages multi-modal artificial intelligence to provide advanced features such as AI-based forensic summarization, music fingerprinting, location detection, and real-time fact-checking. Additionally, it supports a wide array of social platforms including Instagram, TikTok, and YouTube, ensuring a seamless workflow for decoding digital trends.

Key Features

1. AI Forensic Summarization: The application utilizes Gemini 1.5 Pro/Flash algorithms to generate concise and accurate forensic summaries of complex videos, allowing users to grasp key concepts and visual details quickly.
2. Advanced Music Identification: UnReel can transform background noise into identified tracks using Shazam Core Fingerprinting, providing users with official Spotify links and artist credits.
3. Multi-Lens Intelligence (Mind Mapping): Users can visualize video concepts and their interconnections through AI-generated "Intelligence Lenses" (Location, Shopping, and Fact-Check), enhancing their understanding of complex visual topics.
4. Interactive Contextual Chat: This feature enables users to "talk to their Reel," asking specific questions about the video content, transcript, or visual evidence via a persistent RAG-powered chat panel.
5. Multi-Platform Support: The application supports seamless ingestion from Instagram, TikTok, YouTube Shorts, X (Twitter), and LinkedIn, extracting essential content with a high-resilience download pipeline.

6. Forensic Progress Mapping: Users can track the analysis in real-time through a detailed forensic breakdown, categorizing and managing their search filters effectively.
7. Secure Authentication & User Profile: A secure login system powered by Clerk ensures data privacy, while a user profile page provides personalization and tracking of analysis history.

Technical Architecture



The UnReel ecosystem is built on a modular, high-performance stack designed for scale and resilience:

Infrastructure & Deployment

- Frontend (Production): un-reel-app.vercel.app (Deployed on Vercel with global edge caching).
- Backend (Intelligence API): Hugging Face Space: UnReel-API (Deployed as a Dockerized FastAPI application on high-performance cloud hardware).

API Endpoints

The backend exposes a sophisticated RESTful API for deep content interrogation:

- POST /analyze/ - Initiates the multi-lens forensic analysis of a provided URL.
- POST /chat/ - Context-aware AI chat interaction within a specific video's forensic scope.
- GET /history/ - Retrieves user-specific analysis history and previous chat sessions.
- GET /health/ - Real-time system status and service availability monitoring.

The UnReel AI-powered app follows a modular and scalable architecture consisting of several key components:

- Frontend: Developed using Next.js 14+ (App Router), ensuring a high-performance, responsive, and intuitive user interface with Framer Motion animations.
- Backend: Powered by FastAPI (Python), handling heavy media processing, asynchronous tasks, and AI orchestration via Docker containers.
- AI Services: Integrates state-of-the-art multimodal models via Google Gemini for visual reasoning, Shazam Core for music fingerprinting, and Google Search for real-time RAG verification.
- Database: Uses PostgreSQL (SQLAlchemy) for secure data storage and persistent analysis/chat history synchronization.
- Authentication: Implements Clerk-based authentication for multi-factor user security and session management.
- File Handling: Supports sophisticated media extraction using FFmpeg for audio/frame processing and hardened yt-dlp for resilient video acquisition.

Competitive Advantage

Unlike other video analysis tools or basic transcribers, UnReel is specifically designed for forensic content investigation and trend research. Our unique advantage lies in:

- Verified Accuracy: Real-time cross-referencing of video claims with live search results using RAG.
- Deep Music Forensic: Resolving background music even in noisy environments via actual fingerprint hashes.
- Visual Intelligence: Recognizing products, locations, and text inside the video frames in 4K resolution.
- Interaction: The ability to treat video content as a "Living Document" through persistent AI chat.

Future Enhancements

The roadmap includes upcoming features such as:

- Batch Analysis: AI-powered analysis for entire content profiles or hashtag collections.
- Chrome Extension Integration: Allowing users to trigger forensic analysis directly from their social feeds.
- Real-time Collaboration: Enabling multiple researchers to share and annotate forensic reports in real-time.
- E-commerce Direct-Links: Connecting identified products in videos directly with global marketplaces.

Conclusion

This AI-powered forensic video intelligence application is a revolutionary tool for creators and researchers, combining the best of multimodal AI, intuitive design, and investigation-enhancing features. By streamlining the trend-watching and verification process, it empowers users to understand content more efficiently and stay ahead of the digital curve. With continuous improvements and future enhancements, UnReel is set to become an indispensable asset for the professional research and creative community.

Live Deployment : Click Here

GitHub Link of UnReel: Video Intelligence and Analysis Platform